

⚡ WD002 Week 2 — English Worksheet

Topic: Variables and the DOM — Changing the Page Live · **Course:** Web Dev Basics: JavaScript · **Time:** about 45 minutes

This week you reach into the page and change it with code. A **variable** (`let`) stores a piece of information. `document.querySelector()` grabs an element by its `id`, and `.textContent` swaps the words inside it. Put them together and a click can rewrite the screen.

1 · Predict 🧠

Read each piece of code. Before you run it, write what you think will happen.

```
<h1 id="title">Hello</h1>

<script>
  let element = document.querySelector("#title");
  element.textContent = "Goodbye";
</script>
```

What word ends up on the screen — `Hello` or `Goodbye` ? Why?


```
<script>
  let name = "Minjun";
  let greeting = "Hi, " + name + "!";
  console.log(greeting);
</script>
```

What full sentence does the console show? What is the `+` doing here?


```
<h1 id="title">Start</h1>
<button onclick="change()">Go</button>

<script>
  function change() {
    let element = document.querySelector("#title");
    element.textContent = "Changed!";
  }
</script>
```

Before the button is clicked, what does the title say? After one click?


```
<script>
  let count = 5;
  count = count + 1;
  console.log(count);
</script>
```

A variable can be changed after it is made. What number prints?

2 · Spot the Bug 🐛

Each block was meant to do something, but it is broken. Read what it is **supposed** to do, then rewrite it so it works. After that, explain why the original was wrong and why your fix works.

Bug A — This should change the heading text to `Welcome`. Right now nothing changes on the page.

```
<h1 id="title">Hello</h1>

<script>
  let element = document.querySelector("title");
  element.textContent = "Welcome";
</script>
```

Hint: to select something by its `id`, the selector needs a special symbol in front.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — This should show `Hi, Sora!` on the page. Right now it shows the words `Hi, name!` exactly.

```
<h1 id="out"></h1>

<script>
  let name = "Sora";
  document.querySelector("#out").textContent = "Hi, name!";
</script>
```

Hint: the variable is stuck inside the quotes, so it is treated as plain text. Pull it out and join it with `+`.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — Clicking should change the heading. Right now you get an error because the element cannot be found.

```
<h1 id="message">Hi</h1>
<button onclick="update()">Update</button>

<script>
  function update() {
    let element = document.querySelector("#message");
    element.textContent = "Updated!";
  }
</script>
```

Hint: compare the `id` in the HTML with the one in the selector, letter by letter.

Write the fixed code:

Why was it wrong? Why does your fix work?

one button changes 3 or more things

- +

different

photo or video

id

...

on changes ...

...

...

d ...

4 · Record Your Walkthrough 🎥

Take a video on your phone (or a parent's phone) while your page runs. Talk through it like you are teaching someone who has never seen it. Try to use these words: **variable**, **DOM**, **querySelector**, **textContent**, **id**.

1. Show the page before clicking.
2. Click the button and show everything change at once.
3. Read one line out loud and say which element it grabs.
4. Show where your variable is used.

Write what you will say in your video. Plan it here before you record — you can read from it while filming.

Submit ✓

Send this worksheet + your walkthrough video to teacher on KakaoTalk.